Avoiding Negative Transfer for Semantic Segmentation of Remote Sensing Images

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***Abstract*— Reducing the feature distribution shift caused by the factor of visual-environment changes, named visual-environment changes (VE-changes), is a hot issue in domain adaptation learning. However, in the semantic segmentation task of remote sensing imageries, besides VE-changes, the change of semantic-scene changes (SS-changes) is another factor raising the domain gap, which brings the label distribution shift. For example, although urban and rural share the same landcover label, there is still a gap in label distribution. If there is little relation that can be found in neither feature nor label space, forcibly adapting to a new domain could have a high risk of negative transfer. Hence, we propose a new Transitive Domain Adaptation method for Remote Sensing (TDARS) images. First, we introduce an intermediate domain to enlarge the relation between the given source and target domains. Second, we learn from primary and nonprimary confident classes to increase the likelihood of transferring valuable information. As a result, TDARS enables the given source and target domains to be connected through the selected intermediate domain and performs effective knowledge transfer among all domains. The proposed method is evaluated on three domain adaptation datasets of remote sensing images. Extensive experiments show that the approach can effectively handle the domain shift problem from remote sensing images compared to other state-of-the-art domain adaptation methods.**

***Index Terms*— Domain adaptation, remote sensing, semantic segmentation.**

I. Introduction

IN MANY real-world applications, we wish to apply models learned on large-scale labeled datasets (the source domain) to a newly unlabeled dataset (the target domain). Unfortunately, learned models may not generalize well to the target domain because of the domain shift problem. To overcome this problem, unsupervised domain adaptation (UDA) [32] with the aim of closing the gap between the source domain and the target domain has received increasing interest.

In the field of semantic segmentation, as illustrated in Fig. 1, the domain shift problem is generally caused by two major factors, i.e., feature distribution shift caused by VE-changes and label distribution shift caused by semantic-scene changes (SS-changes). For most typical applications in the field of computer vision, VE-changes are the leading factor. For example, it can be found that there exists a large visual domain gap between different autonomous driving datasets since images are generally collected under different weather and illumination conditions. However, SS-changes are seldom. In many real-world semantic segmentation tasks of remote sensing imageries, data from the source domain and the target domain are collected at time series with seasonal variations of visual environment and acquired from different cities and countries at different latitudes and longitudes with various semantic scenes. The domain shift problem from remote sensing images is more serious because it is simultaneously caused by both two factors. First, analysis of the time series of remote sensing data shows that there exist seasonal VE-changes [46]. Such VE-changes are driven by seasonal variations in temperature and rainfall interactions impacting landcover spectral reflectance [3]. Various visual environments from different seasons lead to a gap in visual feature distribution across domains. Second, the common phenomenon of SS-changes among remote sensing data can be known in two aspects. From the point of physical geography, data from different latitudes and longitudes reveal infinite SS-changes composed of limited landcover labels that are driven by climatic variability [13], [30], [42], [50]. In addition, from the perspective of human geography, landcover modifications by human activities [10], [21], [31], [33], e.g., urbanization, in different cities and countries highlight frequently SS-changes. These SS-changes caused by climatic variability and human activities present the instability of landcover patterns and the disagreement of label distributions across domains. As a result, although the samples from different domains contain the same landcover label, there still exists a gap in label distribution between the source domain and the target domain.

In UDA, the source domain has well-labeled training data, yet the data from the target domain are totally unlabeled. An important requirement for successful UDA is that the related information between the source domain and the target domain overrides the irrelevant information. This relation usually can be expressed in the similarity of feature or label space. To transfer knowledge into the target domain, one common solution is to utilize the pretrained source model to select quality-related target samples with high confidence scores and assign them class labels as the transferred samples for target feature learning. However, if no direct relation can be found

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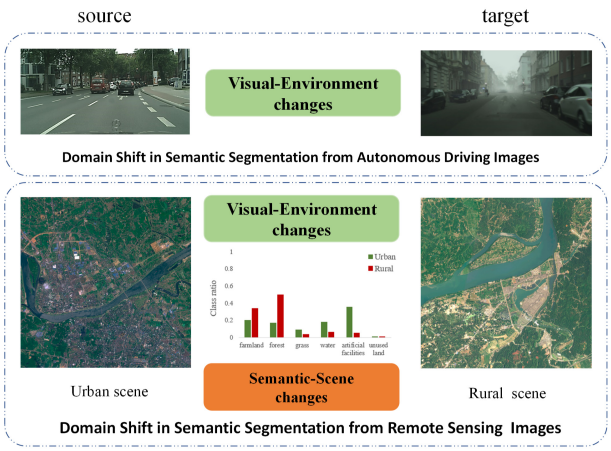


Fig. 1. Comparison of domain shift problem in semantic segmentation from automated driving images and remote sensing images. Different from less changeable scenes in the viewpoint of autonomous vehicles, SS-changes are more common in remote sensing images. Therefore, the domain shift problem from remote sensing images is more serious because it is simultaneously caused by both VE-changes and SS-changes.

in neither feature nor label space, forcibly adapting to a new domain may not work. In the condition of feature distribution shift, features that belong to the same label between the target domain and the source domain are not equal [5]. The major labels in the target domain are inconsistent with those in the source domain because of label distribution shift [44]. Hence, under the combined impact of feature distribution shift and label distribution shift, features that belong to the major labels in the target domain are induced to approach mismatched features in the source domain. In other words, target samples with high confidence scores can still have a high probability of being misclassified. Such misclassified noise will affect the quality of transferred samples. The introduced noise increases as the number of iterations increases during the UDA training, resulting in the negative transfer where the noise information overrides the benefit of adding more transferred samples. Existing majority approaches for UDA are based on feature-distribution alignment [5], [15], [35], [43] or label-distribution alignment [53], [54], [56], [57]. Feature-distribution-alignment UDA methods focus on invariant feature learning between source and target domains through metric learning [16], [37], adversarial learning [9], [15], [43], or image translation [40], [41]. Label-distribution-alignment UDA methods reduce domain gap from label space, which usually minimizes the domain shift by utilizing confidence pseudolabels through self-training [53], [54], [56]. However, these methods take no consideration of the risk of negative transfer under the combined impact of feature distribution shift and label distribution shift. Hence, existing methods for UDA may struggle to handle the domain shift problem from remote sensing images.

In this article, we propose a new Transitive Domain Adaptation method for Remote Sensing (TDARS) images. To decrease the risk of negative transfer and achieve a smooth domain adaptation, we have the following two main issues that should be considered: 1) how to enlarge the relation between the given source and target domains and 2) how to do effective knowledge transfer among domains. To solve the first issue, we propose to construct an intermediate domain to provide more related information between the source and target domains. As VE-changes and SS-changes are mainly corresponding to the variation in temporal and spatial, respectively, we approximately consider the consistency of environment and semantics as temporal consistency and spatial consistency, respectively. That is, we can easily construct an intermediate domain that keeps the visual environment similar to the source domain and remains semantic scene unchanged with the target domain from available time-series remote sensing data [46], referred to as single factor changes between the intermediate domain and source (target) domain. To solve the second issue, considering that learning from misclassified transferred samples can affect learning reliable knowledge, we propose to learn from both primary and nonprimary confident classes to increase the likelihood of transferring valuable information among domains. For each pixel of an image in new domains, the output probability generated by pretrained network parameters belongs to a C-dimensional score space. One class with the most confident score is referred to as the primary confident class. Nonprimary confident classes refer to a set of classes except the primary class. We first give a negative label [19], [20] to one class randomly selected from nonprimary classes, which implies that "a pixel does not belong to this label," to train the adapted network to prevent the model from overfitting to misclassified transferred samples. After that, the adapted network is expected to eliminate some misclassified transferred samples. We assign the filtered transferred samples positive pseudolabel through the primary confident class to further transfer beneficial information. As a result, TDARS enables to not only connect the given source and target domains through the selected intermediate domain but also perform effective knowledge transfer among domains.

We validate the proposed method for semantic segmentation of remote sensing images in three domain adaptation datasets. We also compare the proposed approach with state-of-the-art domain adaptation methods and conduct ablation studies for a detailed analysis. The main contributions of this article are summarized as follows.

1) We propose a TDARS image, which enables the given source and target domains to be connected through an intermediate domain and performs effective knowledge transfer among all domains. The adapted network, thereby, decreases the risk of negative transfer under the combined impact of feature distribution shift and label distribution shift.
2) We adopt temporal consistency and spatial consistency to approximate the consistency of environment and semantic, respectively, which makes the selection of intermediate domain easier to keep a single factor change with the source (target) domain. Note that the selected images from the intermediate domain are totally unlabeled.
3) We experimentally investigate the risk of negative transfer on semantic segmentation of remote sensing images under the existing majority UDA methods that take

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no consideration of the combined impact of feature distribution shift and label distribution shift.

II. Related Work

A. Unsupervised Domain Adaptation

1) Feature-Distribution-Alignment UDA: Feature distribution alignment UDA methods aim at invariant feature learning between source and target domains. These methods can be taxonomically divided into three categories. The first category is the discrepancy-based approach that utilizes different metric learning modes to push feature distributions from different domains close to each other. Maximum mean discrepancy (MMD) is commonly used in various methods [7], [23], [25], [26], [27], [48]. Other discrepancy-based methods include subspace alignment [14], $\mathcal{H}$ divergence [1], geodesic flow kernel [16], RankSVM [28], and correlation alignment [37]. The second category is the adversarial-based approach. With the success of generative adversarial networks (GANs) [17], a series of works [9], [15], [22], [45] focus on feature distribution matching by adopting adversarial learning. Ganin and Lempitsky [15] and Tzeng *et al.* [45] proposed a domain adversarial neural network to encourage the domain confusion by an adversarial objective. Li *et al.* [22] integrated GANs into the model predictor and preserve the semantic information to enhance the performance of the predictor. Cui *et al.* [9] presented a gradually vanishing bridge framework for balanced adversarial adaptation. The third category is the image-translation-based approach, which is motivated by the translation of visual texture across domains and utilizing translated images as augmented training data to help the model learn domain-invariant features. Choi *et al.* [8] utilize a target-guided generator to produce images conditioned on style information extracted from the target domain.

2) Label-Distribution-Alignment UDA: The percentage of each category should not be expected to be similar between source and target domains. There also exists the divergence between known source and unknown target label distribution. Label distribution alignment methods focus on reducing the domain gap from label space. Hsu *et al.* [18] utilized label and structural information within and across domains to reduce domain discrepancy from imbalanced data. Yan *et al.* [52] exploited the class prior probability on the source and target domains by introducing class-specific auxiliary weights into MMD. Ding *et al.* [12] constructed two coupled neural networks and designed a weighted classwise adaptation term to fight off the label distribution mismatch.

B. UDA for Semantic Segmentation

1) Semantic Segmentation of Natural Images: For most typical applications in the field of computer vision, semantic segmentation faces challenges from the variance in illumination, viewpoints, instance appearance, weather conditions, and so on, which could mainly sum up to the changes of the visual environment. The UDA methods for semantic segmentation can also be taxonomically divided into the feature-distribution-alignment approach and the label-distribution-alignment approach. A series of works based on feature alignment typically consider the UDA for semantic segmentation as a structural adaptation problem. Chang *et al.* [5] proposed to disentangle images into domain-invariant structure and domain-specific texture representations for better image translation across domains. Tsai *et al.* [43] proposed to learn adversarial domain-invariant features from structured similarity across domains, which is then extended by introducing entropy minimization objective [47]. Saito *et al.* [35] maximized the discrepancy between two classifiers and minimized the feature extractor part of the network to align distributions across domains. On the other hand, label distribution alignment methods attempt to solve UDA for semantic segmentation in a self-training manner. Zou *et al.* [56] adopted self-training to iteratively generate pseudolabels on the target data and then utilize these pseudolabels to retrain the network. Zou *et al.* [57] introduced confidence regularization terms to reduce noise in pseudolabels. Zhang *et al.* [54] generated pseudolabels based on categorical centroids. Zhang *et al.* [53] considered prototypical context estimated in the target domain to enforce the alignment at the label level. However, all these literature works overlook the risk of negative transfer under the combined impact of feature distribution shift and label distribution shift. Hence, these methods may fail to solve the domain shift problem from remote sensing images, which is caused by both two factors.

2) Semantic Segmentation of Remote Sensing Images: Compared to natural images, the UDA for remote sensing images suffers a more serious domain shift problem caused by both feature distribution shift and label distribution shift, which is ascribed to frequent VE-changes and SS-changes. Earth observation data in different cities and countries are usually acquired from various geographical environments. For example, Wuhan sits at the confluence of the Han River with abundant lakes, whereas Xiangtan is located on the lower reaches of the Xiang River and surrounded by low mountains and hills. On the one hand, despite limited landcover labels, varying degrees of terrains and climates can cause various SS-changes [13], [30]. On the other hand, human activities affect landcover modifications, leading to frequent SS-changes [10], [31]. Driven by the power of deep convolutional neural networks, domain adaptation of semantic segmentation has also raised increasing attention in remote sensing. In early works, many attempts have been made to introduce UDA techniques to overcome the domain shift from remote sensing images. Bruzzone and Prieto [4] proposed an expectation-maximization method to retrain classifiers for pixel-level adaptation of multitemporal remote sensing images. Deng *et al.* [11] simultaneously learn knowledge from the source domain and aligns the feature extracted from both domains. Ma *et al.* [29] proposed a covariance alignment algorithm for UDA of remote sensing images. Recently, to overcome the domain gap from various visual environments, Tasar *et al.* [40] proposed an image translation method to perform style transfer between different remote sensing images. It first utilizes an adaptive instance normalization to train a data augmentor to implement the translation of visual texture across domains, which is based on a unique and constant style code for each domain. Then, the augmented images are

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further used to train a classifier for domain-invariant feature learning. To overcome the domain gap from the change of semantic scenes, Benjdira *et al.* [2] proposed a method where CycleGAN [55] is applied in the cross-semantic-scene domain adaptation process. Shi *et al.* [36] constructed a JPRNet with two branches that are based on CycleGAN to cooperatively reduce domain shift from cross-semantic-scene images at the pixel level and the representation level. Li *et al.* [24] proposed to impose multiple weakly supervised constraints on the deep semantic segmentation network training process for cross-semantic-scene domain adaptation. However, these approaches are still limited for the adaptation among remote sensing images with significant VE-changes and SS-changes. Therefore, how to solve the domain shift problem from remote sensing images for wide practical applications remains to be studied.

C. Transitive Transfer Learning

An essential assumption for transfer learning is that the source domain and the target domain have a close relation in the form of related instances, features, or models. Transferring knowledge between two domains without direct relation forcibly will not work. It could lead to even worse performance than "source-only" algorithms in the target domain, which is called "negative transfer" [34]. To overcome this limitation of traditional transfer learning, Tan *et al.* [38] proposed transitive transfer learning (TTL) that is inspired by the human ability to conduct transitive inference. The assumption of TTL is that, although the source and target domains have few common factors, they could be bridged by intermediate domains through some underlying shared factors.

Early work [38] was targeted at utilizing intermediate data to transfer knowledge between text data in the source domain and the image data in the target domain. Wu *et al.* [51] develop this method for online heterogeneous transfer learning, which transits knowledge from a heterogeneous source domain to a target domain through intermediate data. It takes co-occurrence data, a set of text-image pairs, as the intermediate domain because unlabeled co-occurrence data are cheap and easily collected from web pages or social networks. Tan *et al.* [39] select subsets from a mixture of intermediate domains to bridge source and target domains.

However, the definition and selection of the intermediate domain are problem-specific. Existing TTL algorithms mostly focus on the knowledge transfer from text data to image data or the domain transfer among images that embody one single nature object. Therefore, how to define the intermediate domain from various remote sensing images is required to be studied in depth.

III. Approach

A. Why Traditional DA Methods Fail to Solve the Domain Shift Problem From Satellite Images

Given a source domain $\mathcal{D}_S = \{(x_S^i, y_S^i)\}_{i=1}^{N_S}$ of N_S labeled examples, where each sample $x_S^i \in \mathbb{R}^{(H\times W\times 3)}$ is a $(H \times W \times 3)$ remote sensing image along with associated C-class segmentation map $y_S^i \in [1, C]^{(H\times W)}$. Given a target domain $\mathcal{D}_T = \{(x_T^i)\}_{i=1}^{N_T}$ of N_T unlabeled samples with the same label set.

Let $P(x_S)$ and $P(y_S|x_S)$ be the marginal and conditional distributions of the source domain. Also, let $P(x_T)$ and $P(y_T|x_T)$ be the parallel definitions of the target domain. Given a domain divergence measure $\mathrm{div}(\cdot;\cdot)$, the domain shift problem can be formulated as

$$\mathrm{div}(P(x_S)P(y_S|x_S),\, P(x_T)P(y_T|x_T)). \tag{1}$$

The domain shift is generally caused by two major factors, feature distribution shift and label distribution shift, which could be ascribed to two phenomena, VE-changes and SS-changes, respectively. VE-changes can cause inconsistency in feature space, i.e., $P(x_S) \neq P(x_T)$. SS-changes lead to a mismatch in label space, i.e., $P(y_S|x_S) \neq P(y_T|x_T)$.

For most typical applications in the field of computer vision, due to seldom SS-changes, the impact of SS-changes is often overlooked. For example, under the hypothesis of $P(y_S|x_S) \approx P(y_T|x_T)$, a domain adaptation task from autonomous driving datasets $\mathcal{M}_a$ from the source domain S to the target domain T can be approximatively defined as

$$\mathcal{M}_a : \mathbb{E}_{(x_S)\sim\mathcal{D}_S} P(x_S) \to \mathbb{E}_{(x_T)\sim\mathcal{D}_T} P(x_T). \tag{2}$$

However, both VE-changes and SS-changes are frequent in the domain adaptation task from remote sensing images, which can be defined as

$$\mathcal{M}_{rs} : \mathbb{E}_{(x_S,y_S)\sim\mathcal{D}_S}(P(x_S)P(y_S|x_S)) \to \mathbb{E}_{(x_T,y_T)\sim\mathcal{D}_T}(P(x_T)P(y_T|x_T)). \tag{3}$$

When samples from source and target domains are less related in neither feature space, i.e., $P(x_S) \neq P(x_T)$, nor label space, i.e., $P(y_S|x_S) \neq P(y_T|x_T)$, features in the target domain are induced to approach semantic-mismatched features in the source domain, leading to a high risk of negative transfer [49].

Most existing UDA methods [15], [40], [47], [53] pay less attention to the phenomenon of SS-changes. Because of this, these methods take no consideration of the negative transfer risk under the combined impact of feature distribution shift and label distribution shift, which severely restrict the application scenarios and struggle to handle the domain shift problem on $\mathcal{M}_{rs}$. Hence, we need a novel perspective to solve the domain shift problem from remote sensing images.

B. Transitive Domain Adaptation for Remote Sensing

First, an intermediate domain that enlarges the relation between the given source and target domains has been defined and selected. Second, we propose to exploit valuable information from primary and nonprimary confident classes to effectively transfer knowledge. Finally, the transitive adaptation learning algorithm is proposed to solve the domain shift problem from remote sensing images.

1) Intermediate Domain:

a) Intermediate domain definition: In the semantic segmentation of remote sensing images, the domain shift problem caused by feature distribution shift and label distribution shift is ascribed to VE-changes and SS-changes, respectively, which makes the source domain $\mathcal{D}_S$ and the target domain $\mathcal{D}_T$

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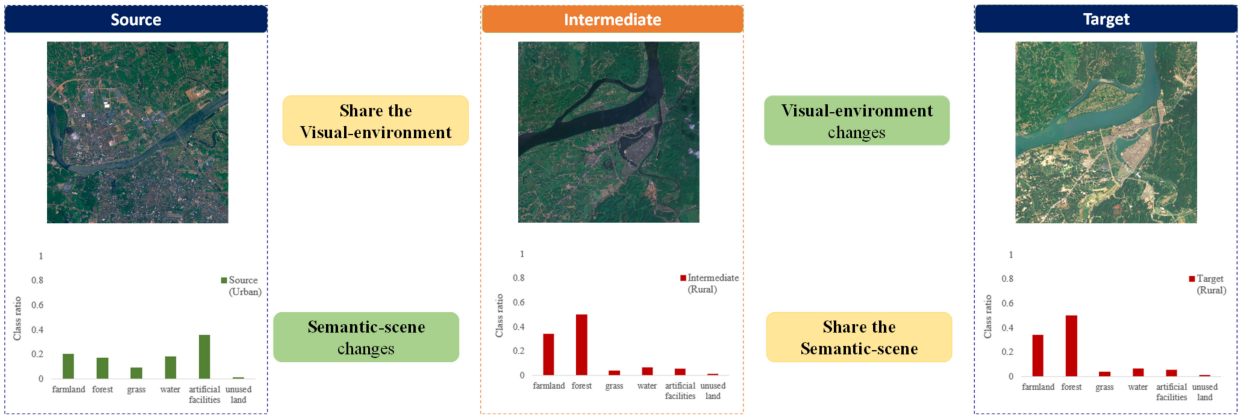


Fig. 2. Intermediate domain to enlarge the relation between the given source and target domains, holding the intermediate domain to share one factor with the source (target) domain. The selected images from the intermediate domain keep the same season as the source domain to share the visual environment. The selected images from the intermediate domain keep the same geospatial area, sensor, and spatial resolution with the target domain to share the semantic scene.

have few related information. We introduce an intermediate domain $\mathcal{D}_I$ to enlarge the relation between the given $\mathcal{D}_S$ and $\mathcal{D}_T$, as shown in Fig. 2. Formally, let $z_S^{\{\beta_S,\sigma_S\}}$ and $z_T^{\{\beta_T,\sigma_T\}}$ be the set of latent factors in source and target domains, where β and σ are the latent factors corresponding to visual feature representation and semantic feature representation, respectively. Given an intermediate domain $\mathcal{D}_I = \{(x_I^i)\}_{i=1}^{N_I}$ of N_I unlabeled samples with a set of latent factors $z_I^{\{\beta_I,\sigma_I\}}$, which keeps similar visual feature representations of the source domain, i.e., $\beta_I \approx \beta_S$, and remains consistent semantic feature representations of the target domain, i.e., $\sigma_I \approx \sigma_T$. Hence, the selected intermediate domain $z_I^{\{\beta_I,\sigma_I\}} \approx z_I^{\{\beta_S,\sigma_T\}}$ will satisfy both

$$\mathrm{div}\left(z_S^{\{\beta_S,\sigma_S\}}, z_I^{\{\beta_S,\sigma_T\}}\right) < \mathrm{div}\left(z_S^{\{\beta_S,\sigma_S\}}, z_T^{\{\beta_T,\sigma_T\}}\right) \tag{4}$$

and

$$\mathrm{div}\left(z_I^{\{\beta_S,\sigma_T\}}, z_T^{\{\beta_T,\sigma_T\}}\right) < \mathrm{div}\left(z_S^{\{\beta_S,\sigma_S\}}, z_T^{\{\beta_T,\sigma_T\}}\right). \tag{5}$$

Given the feature representation function $\mathcal{F}(\cdot)$, by introducing the selected intermediate domain, the domain adaptation task for remote sensing $\mathcal{M}_{rs}$ can be rewritten as

$$\begin{aligned}\mathcal{M}_{rs} : \mathbb{E}(P(\mathcal{F}(x_S) \mid z_S^{\{\beta_S,\sigma_S\}}) \\ \rightarrow \mathbb{E}(P(\mathcal{F}(x_I) \mid z_I^{\{\beta_S,\sigma_T\}}) \\ \rightarrow \mathbb{E}(P(\mathcal{F}(x_T) \mid z_T^{\{\beta_T,\sigma_T\}}).\end{aligned} \tag{6}$$

The intermediate domain is able to strengthen semantic correlation and shrink feature distance between source and target domains.

b) Intermediate domain selection: Spatial and temporal are two important factors in a data query that describes the selection of remote sensing images. First, VE-changes are mainly caused by the variation in temporal. Data from different periods of remote sensing sensors reveal seasonal variations in landcover spectral reflectance, leading to visual variability [47]. Second, SS-changes are primarily caused by the variation in spatial. Geographical landscapes in remote sensing images present significant divergence in different latitudes and longitudes even if they were acquired at the same time [33]. Hence, in this article, we approximately consider the consistency of the visual environment and the consistency of the semantic scene as temporal consistency and spatial consistency, respectively. We summarize temporal consistency and spatial consistency as follows.

1) *Temporal Consistency:* We assume that, if the images come from the same season, they possess temporal consistency. In the analyses of multiyear time-series images, there exist patterns of seasonal regularity in visual appearance. On the one hand, the VE-changes between different seasons often show abrupt and nonlinear variation. For example, due to the seasonal changes in temperature and rainfall interactions impacting plant phenology, not all crops are suitable to grow in every season. Therefore, "Farmland" usually has different visual appearances in different seasons because different crops should be planted according to the seasonal change. On the other hand, the VE-changes within each season occur in a progressive way. It means that, if two images are acquired from the same season, they are more likely to have a similar visual environment. Hence, we select the images that share the same season with the source domain to approximately satisfy the character of a similar visual environment.
2) *Spatial Consistency:* Time-series remote sensing data are taken from the same location at different times through various remote sensing sensors and spatial resolutions. We hypothesize that, if images are collected at the same geospatial area from the same sensor at a unified spatial resolution, they possess spatial consistency. On the one hand, various spatial resolutions have a significant impact on clustered spatial patterns, leading to the SS-changes. For example, individual buildings and cars can be detected from very high-resolution images, while coarse-resolution images are only enabled to distinguish the boundary of a province or country. On the other hand, the SS-changes are also highlighted between images captured from various remote sensing sensors

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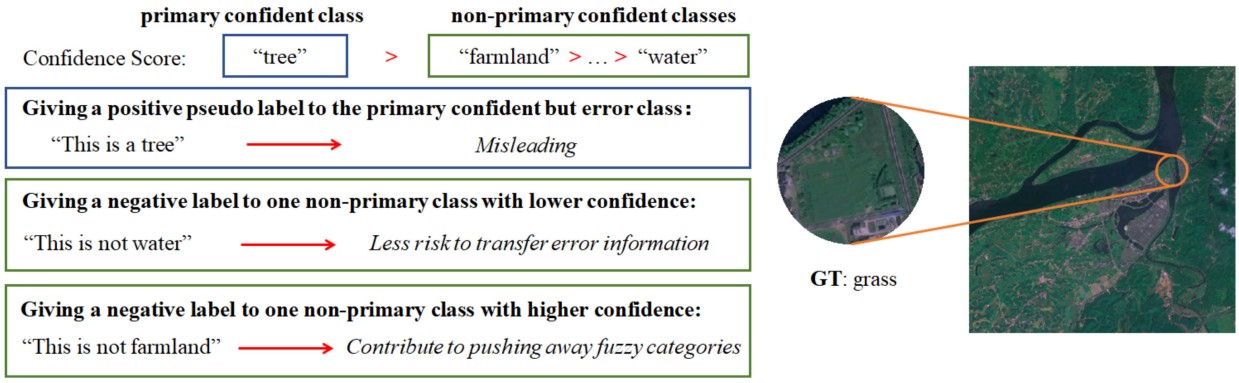


Fig. 3. Knowledge transfer from nonprimary confident classes. Compared to misleading adapted models through error pseudolabels, giving a negative label to one class randomly selected from nonprimary classes may decrease the risk of transferring error information and push away fuzzy categories.

because different sensors record different characteristics of the land surface. For example, multispectral sensors respond to reflected light or temperature, whereas radar sensors respond to orientation, surfaces, and volume of objects. Hence, we select the images that share the same geospatial area, sensor, and spatial resolution with the target domain to satisfy the character of a closer semantic scene.

To this end, we introduce an intermediate domain to enlarge the relation between the given source and target domains. The selected images from the intermediate domain keep the same season as the source domain to share the visual environment. The selected images from the intermediate domain keep the same geospatial area, sensor, and spatial resolution with the target domain to share the semantic scene.

2) Knowledge Transfer via Primary and Nonprimary Confident Classes: One common solution to transfer knowledge is utilizing a pretrained model to select quality samples from new domains with high confidence scores and assign them pseudolabels as transferred samples. Let θ_{pre} and θ denote the parameters of the pretrained network and the adapted network, respectively. $f(x;\theta)$ denotes the map function from the input space to the C-dimensional score space. The loss function of the above pixelwise knowledge transfer solution is given as follows:

$$\mathcal{L}_{\text{pixel}} = -\frac{1}{W \times H}\sum_{i \in R} \max\left(f\left(x^i;\theta_{\text{pre}}\right) > \gamma\right)\hat{y}^i \times \log f\left(x^i;\theta\right)\Big) \quad (7)$$

where γ denotes the confidence threshold, $\hat{y}^i$ denotes the pseudolabels corresponding to the ith sample x^i, and $R = \{1, 2, \ldots, W \times H\}$ denotes all pixels. However, simply maximizing the confidence score of pseudolabels may fail to guarantee a positive adaptation because transferred samples with high confidence scores can still have a probability of being misclassified. Along with increasing iterations, learning from misclassified transferred samples introduces increasing noise information, resulting in the negative transfer [49] where the noise information overrides the benefit of adding more transferred samples.

For each pixel of a sample from new domains, the output probability p generated by pretrained network parameters θ_{pre} belongs to a C-dimensional score space, i.e., $p = [\ldots, k, \ldots]^T \in \mathbb{R}^C$, where k is the most confident score. One class with the most confident score is referred to as the primary confident class and nonprimary classes refer to a set of classes except the primary class. We consider exploiting valuable information from nonprimary confident classes to alleviate the risk of overfitting misclassified samples. On the one hand, samples far away from decision boundaries are more probably to classify accurately. Thus, taking one nonprimary class with lower confidence as a negative label have less risk to transfer error information, which implies that "this input sample does not belong to this label." On the other hand, fuzzy categories tend to be near the decision boundaries. That is, taking one nonprimary class with a little higher confidence as a negative label may provide the clue to pushing away fuzzy categories. For example, as shown in Fig. 3, when training merely from pseudolabels, if the source model predicts an input of grass as the label "tree," the adapted model will be misled by this wrong information. However, we randomly generate two complementary negative labels from nonprimary classes, for example, "not farmland" and "not water." Training adapted model with negative labels to acknowledge that this input is not farmland or not water is transferring valuable information. Moreover, in this case, if the confidence score of "farmland" is higher than "water," which implies that "farmland" is more easily confused with the ground truth. Training with "not farmland" may contribute to pushing away fuzzy categories. As a result, we give a negative label to one class randomly selected from nonprimary classes, which may prevent the adapted model from overfitting to misclassified transferred samples.

Given pixelwise confidence score p of a sample from new domains, $\hat{y}$ denotes the pixelwise primary confident class, namely, pseudolabels. $\bar{\bar{\mathcal{Y}}}$ and $\bar{\mathcal{Y}}$ denote the set of the higher and lower confidence score in nonprimary categories, respectively. $\bar{\mathcal{Y}} < \bar{\bar{\mathcal{Y}}} < \{\hat{y}\}$ is in an ascending sort order. Let the number of fuzzy categories be $L < C$. Then, the lengths of $\bar{\bar{\mathcal{Y}}}$ and $\bar{\mathcal{Y}}$ are L and $C - L - 1$, respectively. In this study, we set L to 2. The generation of the fuzzy negative label (a higher possibility to distinguish fuzzy categories) $\bar{\bar{y}}$ and less-risk negative label

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Algorithm 1 Negative Label Generation

Input: the training sample x from new domains, the set of pre-trained network parameters θ_{pre}

1: **while** iteration **do**
2: *pixel-wise ascending sort from confidence score* $f(x;\theta_{pre}) : \bar{\mathcal{Y}} < \bar{\bar{\mathcal{Y}}} < \{\hat{y}\}$
3: $\bar{y}$ = *Randomly select from* $\{1,\ldots,C\}\backslash(\{\hat{y}\}\cup\bar{\bar{\mathcal{Y}}})$
4: $\bar{\bar{y}}$ = *Randomly select from* $\{1,\ldots,C\}\backslash(\{\hat{y}\}\cup\bar{\mathcal{Y}})$

Output: less-risk negative label $\bar{y}$, fuzzy negative label $\bar{\bar{y}}$

(a higher probability to transfer correct information) $\bar{y}$ is shown in Algorithm 1.

Equation 7 is suitable for pseudolabels, which is optimizing the value corresponding to the given label as 1 ($p_y \to 1$). However, negative labels differ from pseudolabels as they are expected to be optimized the corresponding output probability to reach 0 ($p_{\bar{y}} \to 0$, $p_{\bar{\bar{y}}} \to 0$). Therefore, the loss functions of pixelwise knowledge transfer from less-risk and fuzzy negative labels are, respectively, defined as

$$\mathcal{L}_{\text{rNeg}} = -\frac{1}{W\times H}\sum_{i\in R}\left(\bar{y}^i \log\left(1-p^i\right)\right) \tag{8}$$

$$\mathcal{L}_{\text{fNeg}} = -\frac{1}{W\times H}\sum_{i\in R}\left(\bar{\bar{y}}^i \log\left(1-p^i\right)\right) \tag{9}$$

where $p^i = f(x^i,\theta)$, and $R = \{1,2,\ldots,W\times H\}$ denotes all pixels. The overall loss function of pixelwise knowledge transfer from negative labels is defined as

$$\mathcal{L}_{\text{Neg}} = \mathcal{L}_{\text{rNeg}} + \alpha * \mathcal{L}_{\text{fNeg}} \tag{10}$$

where α is a balance weight that trades off the effect of less-risk and fuzzy negative labels. After learning through the proposed negative labels, the adapted network is expected to eliminate some misclassified transferred samples. We take the primary confident class as positive pseudolabels for the filtered transferred samples to further learn beneficial information. The loss function of pixelwise knowledge transfer from positive pseudolabels is defined as

$$\mathcal{L}_{\text{Pos}} = -\frac{1}{W\times H}\sum_{i\in R}\max\left(p^i > \gamma\right)\hat{y}^i \log\left(p^i\right)\Big) \tag{11}$$

where $p^i = f(x^i,\theta)$, γ denotes the confidence threshold, $\hat{y}^i$ denotes the pseudolabel corresponding to the ith sample x^i, and $R = \{1,2,\ldots,W\times H\}$ denotes all pixels.

3) Learning Algorithm: To decrease the risk of negative transfer, the proposed method enlarges the relation between the given source and target domains through the selected intermediate domain and does effective knowledge transfer by exploiting the information from primary and nonprimary confident classes. As we discussed in Section III-B2, giving a negative label to one class randomly selected from nonprimary classes may prevent the adapted model from overfitting to misclassified transferred samples and giving positive pseudolabels to primary class for the filtered transferred samples to learn beneficial label information.

The parameters of source network θ_S^* are obtained by minimizing the cross-entropy loss on source labeled data

$$\mathcal{L}_{\text{src}} = -\frac{1}{W\times H}\sum_{i\in R}\left(y_S^i \log f\left(x_S^i,\theta_S\right)\right). \tag{12}$$

Let $\bar{y}_I, \bar{\bar{y}}_I \in [1,C]^{H\times W}$ denote the less-risk and fuzzy negative labels that are generated from source network θ_S^* for intermediate data. The loss functions of pixelwise knowledge transfer from less-risk and fuzzy negative labels for the optimization of intermediate network θ_I^* are defined as

$$\mathcal{L}_{\text{rNeg}-I} = -\frac{1}{W\times H}\sum_{i\in R}\left(\bar{y}_I^i \log\left(1-p_I^i\right)\right) \tag{13}$$

$$\mathcal{L}_{\text{fNeg}-I} = -\frac{1}{W\times H}\sum_{i\in R}\left(\bar{\bar{y}}_I^i \log\left(1-p_I^i\right)\right) \tag{14}$$

where $p_I^i = f(x_I^i,\theta_I)$, and $R = \{1,2,\ldots,W\times H\}$ denotes all pixels. Hence, the overall loss function of pixelwise knowledge transfer from negative labels for the optimization of intermediate network θ_I^* is defined as

$$\mathcal{L}_{\text{Neg}-I} = \mathcal{L}_{\text{rNeg}-I} + \alpha * \mathcal{L}_{\text{fNeg}-I} \tag{15}$$

where α is a balance weight that trades off the effect of less-risk and fuzzy negative labels, default as 0.05. After learning through the proposed negative labels, we train the intermediate network through the filtered intermediate samples with pseudolabels having high confidence over γ. In this research, we set γ to 0.5. Let $\hat{y}_I \in [1,C]^{H\times W}$ denotes the positive pseudolabels that are generated from the source network θ_S^* for intermediate data. The loss function of pixelwise knowledge transfer from positive pseudolabels for the optimization of intermediate network θ_I^* is defined as

$$\mathcal{L}_{\text{Pos}-I} = -\frac{1}{W\times H}\sum_{i\in R}\max\left(p_I^i > \gamma\right)\hat{y}_I^i \log\left(p_I^i\right)\Big) \tag{16}$$

where $p_I^i = f(x_I^i,\theta_I)$, and $R = \{1,2,\ldots,W\times H\}$ denotes all pixels.

After knowledge learned from the source network is well transferred to the selected intermediate domain, we take the intermediate network as a new "teacher." Let $\bar{y}_T, \bar{\bar{y}}_T \in [1,C]^{H\times W}$ denote the less-risk and fuzzy negative labels, and $\hat{y}_T \in [1,C]^{H\times W}$ denotes the positive pseudolabels, which are generated from intermediate network θ_I^* for target data. Also, we could get the parallel definitions of pixelwise knowledge transfer loss functions for the optimization of target network θ_T^*

$$\mathcal{L}_{\text{rNeg}-T} = -\frac{1}{W\times H}\sum_{i\in R}\left(\bar{y}_T^i \log\left(1-p_T^i\right)\right) \tag{17}$$

$$\mathcal{L}_{\text{fNeg}-T} = -\frac{1}{W\times H}\sum_{i\in R}\left(\bar{\bar{y}}_T^i \log\left(1-p_T^i\right)\right) \tag{18}$$

$$\mathcal{L}_{\text{Neg}-T} = \mathcal{L}_{\text{rNeg}-T} + \alpha * \mathcal{L}_{\text{fNeg}-T} \tag{19}$$

$$\mathcal{L}_{\text{Pos}-T} = -\frac{1}{W\times H}\sum_{i\in R}\max\left(p_T^i > \gamma\right)\hat{y}_T^i \log\left(p_T^i\right)\Big) \tag{20}$$

where $p_T^i = f(x_T^i,\theta_T)$, and $R = \{1,2,\ldots,W\times H\}$ denotes all pixels. We give the procedure of the proposed learning algorithm in Algorithm 2.

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Algorithm 2 Transitive Adaptation Learning Algorithm

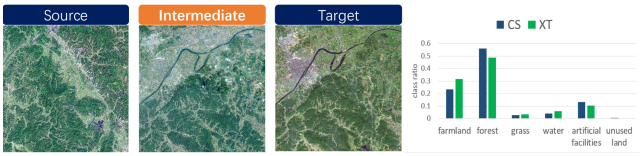
Input: Target and intermediate domains $\mathcal{D}_T$ and $\mathcal{D}_I$; the parameters of target and intermediate network θ_T and θ_I; the parameters of pre-trained source network θ_S^*; and the number of epochs Ep_{max}

while $ep < Ep_{max}$ **do**
  Generate negative labels $\bar{y}_I, \bar{\bar{y}}_I$ from θ_S^*
  Update θ_I according to the rules given in Eq. 13, Eq. 14, Eq. 15
  Generate positive pseudo labels $\hat{y}_I$ from network θ_S^*
  Update θ_I according to the rules given in Eq. 16
while $ep < Ep_{max}$ **do**
  Generate negative labels $\bar{y}_T, \bar{\bar{y}}_T$ from θ_I^*
  Update θ_T according to the rules given in Eq. 17, Eq. 18, Eq. 19
  Generate positive pseudo labels $\hat{y}_T$ from network θ_I^*
  Update θ_T according to the rules given in Eq. 20

Output: the adapted parameters of target network θ_T^* and the predicted results of target samples

TABLE I
DATASET DESCRIPTION

Dataset	Domain	Season	Sensor	Spatial-resolution	256×256 patches
CS	source	Spring	GaoFen-2	2m	43094
WH	source	Spring	GaoFen-2	2m	31272
XT	target	Autumn	GaoFen-2	2m	17963
YY	target	Autumn	GaoFen-2	2m	44795



CS2XT

Fig. 4. Remote sensing images (8192 × 8192) sampled from source and target domains of CS2XT and corresponding label distribution of six categories in the source and target domains. The selected image from the intermediate domain holds the same season with the source domain and shares the same geospatial area, sensor, and spatial resolution with the target domain.

IV. EXPERIMENTS

A. Setups

1) Dataset: Our datasets are RGB remote sensing images with a size of 8192 × 8192 and a spatial resolution of 2 m from GaoFen-2. In our experiments, datasets are acquired over four cities: ChangSha (CS), WuHan (WH), XiangTan (XT), and YiYang (YY). CS and WH datasets are collected in the season of spring, whereas XT and YY datasets are acquired in the season of autumn. All images are cropped to a size of 256 × 256 patches for training or testing without overlap. We were given the annotations for six landcover categories: farmland, forest, grass, water, artificial facilities, and unused land. We conduct three remote sensing adaptation experiments under the setting of both visual environment and SS-changes to evaluate the performance of TDARS and other competitive methods. For example, we take the CS dataset as the source domain and the XT dataset as the target domain, which is named CS2XT. In a similar way, we get another two domain adaptation settings: WH2XT and WH2YY. Table I gives a detailed dataset description.

2) Implementation Details: All experiments are performed on a PC with 3.7-Hz Core E5-1620 CPU, GeForce GTX 1080Ti, and 24-GB RAM. We use DeepLab-v3-plus [6] with ResNet-50 as the semantic segmentation architecture. The datasets are augmented with the random horizontally flip and vertical flip. We set the initial learning rate $LR = 0.02$ for the SGD optimizer and then adjust the learning rate by cosine annealing. The momentum is set to 0.9. To fairly evaluate different domain adaptation methods, we use the same convolutional architecture, augmented strategies, optimizer, and hyperparameters of the optimizer.

B. Adaptation Results of Remote Sensing Images

We compare our proposed approach to four recent domain adaptation methods. MCD [35] and GVBDA [9] are deep domain adaptation models based on feature distribution matching through adversarial learning. ProDA [53] is a label distribution alignment method, which adopts self-training and takes the prototypical context estimated into consideration to correct pseudolabels. DAugNet [40] performs style transfer to transfer remote sensing image texture appearance from the source domain to the target domain and reduces domain shift with augmented fake source data. We reimplement DAugNet and conduct experiments for the rest methods with their open-source codes. We assess the performance by the following evaluation criterion: the frequency weighted intersection over union (FWIoU), the overall accuracy (OA), the average accuracy (AA), the kappa coefficient, and the intersection over union (IoU) for each class.

1) CS2XT: In this case, we use CS as the source city and XT as the target. We can observe that there exists visual-environment discrepancy and label distribution divergence between the source domain and the target domain from Fig. 4. The intermediate domain is collected in the condition that holds the same season with the source domain and shares the same geospatial area, sensor, and spatial resolution with the target domain. Hence, the intermediate domain shrinks domain divergence between source and target domains. Table II depicts the comparison results. Compared with the nonadapted source model (baseline), the performance of MCD and GVBDA, which is based on feature distribution matching through adversarial learning significantly, drops under the domain shift caused by two factors. ProDA considers denoising the pseudolabels, but only utilizing pseudolabels still brings a bad performance. The proposed TDARS achieves competitive or superior performance, outperforming the nonadapted baseline by 2.75% and DAugNet by 1.71% in terms of FWIoU. This can be ascribed to effective knowledge transfer in a transitive learning process. In Fig. 5, we show the qualitative comparison results.

It is noteworthy that class imbalance is a common phenomenon in domain adaptation of remote sensing images,

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TABLE II

COMPARISON RESULTS (%) OF CS2XT IN TERMS OF IoU FOR EACH CLASS, OA, AA, KAPPA, AND FWIPoU. THE BEST IS HIGHLIGHTED IN COLUMN

Method	farmland	forest	grass	water	artificial facilities	unused land	OA	AA	Kappa	FWIoU
src only	67.43	77.30	6.91	55.78	46.29	2.94	80.18	56.63	68.25	67.32
Daugnet	69.15	78.40	6.29	51.48	48.43	**5.56**	80.74	56.40	69.32	68.36
MCD	60.24	72.69	0.05	42.44	35.23	0.00	75.97	54.13	60.44	60.66
GVBDA	66.00	76.65	**8.92**	53.02	47.66	4.06	79.46	56.62	67.27	66.60
ProDA	70.10	77.79	6.73	41.79	37.16	5.10	77.95	51.66	66.33	66.98
TDARS(ours)	**71.38**	**79.96**	3.09	**55.85**	**49.52**	0.00	**82.27**	**59.25**	**71.88**	**70.07**

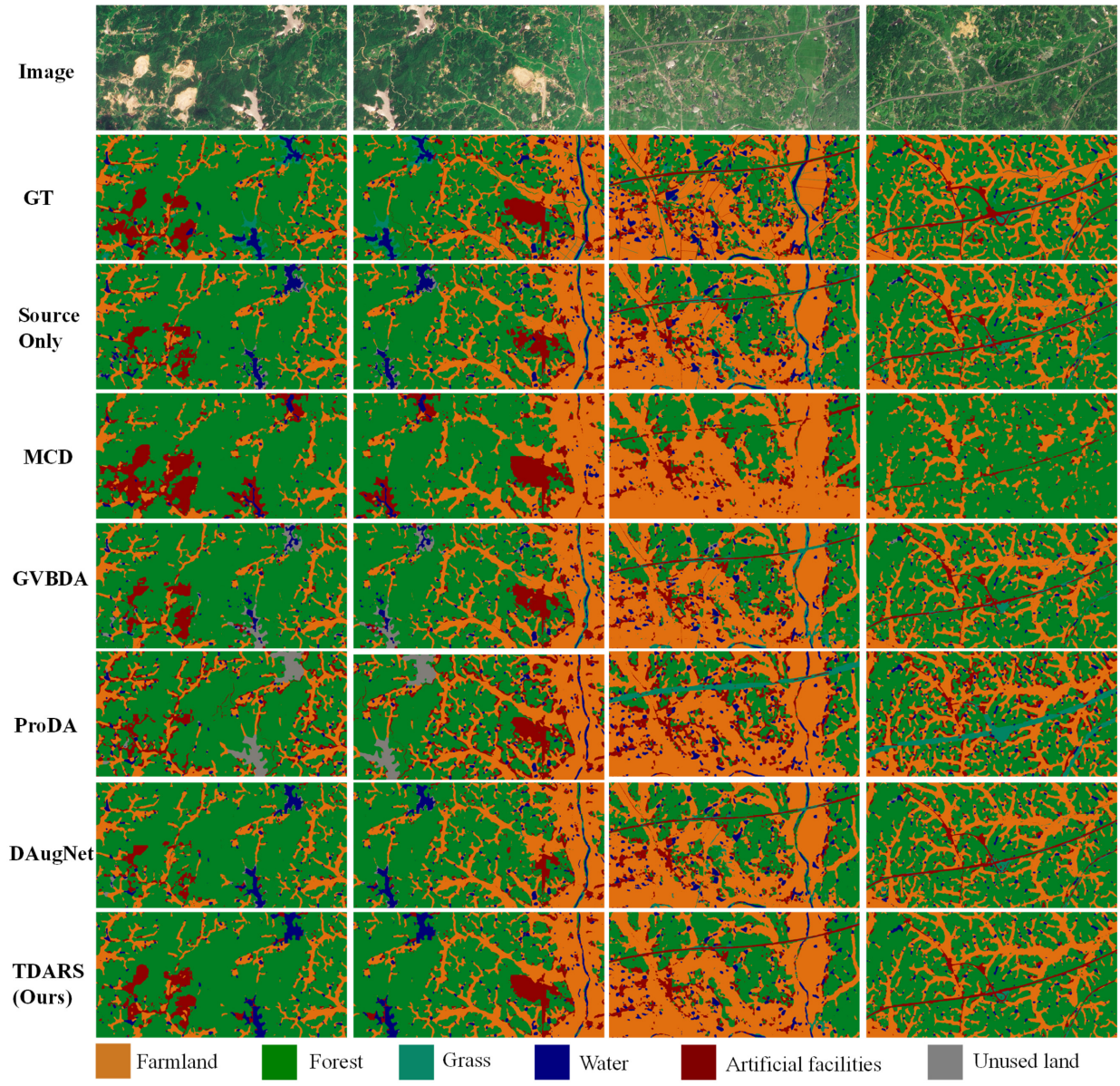


Fig. 5. Qualitative comparisons of adapted segmentation on CS2XT. We splice 32 adapted patches (256 × 256) to a remote sensing image (2048 × 1024).

which causes two problems. The first problem is negative transfer; namely, the overall performance decreases, which has been addressed by the proposed method. Our TDARS reduces the risk of negative transfer and arrives at the best overall performance. The second problem is insufficient adaptation learning on extremely rare classes (e.g., grass and unused land), which remains unresolved. Therefore, the proposed algorithm achieves much lower IoU on extremely rare classes. However, compared methods also have a bad performance in extremely rare classes. When the

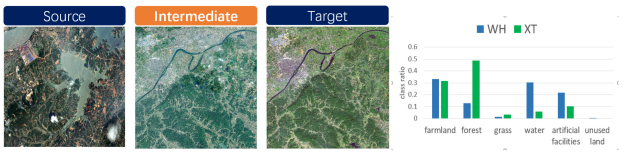


Fig. 6. Remote sensing images (8192 × 8192) sampled from source and target domains of WH2XT and corresponding label distribution of six categories in the source and target domains. The selected image from the intermediate domain holds the same season with the source domain and shares the same geospatial area, sensor, and spatial resolution with the target domain.

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TABLE III
COMPARISON RESULTS (%) OF WH2XT IN TERMS OF IOU FOR EACH CLASS, OA, AA, KAPPA, AND FWIOU. THE BEST IS HIGHLIGHTED IN COLUMN

Method	farmland	forest	grass	water	artificial facilities	unused land	OA	AA	Kappa	FWIoU
src only	42.42	47.75	0.00	16.28	34.61	0.06	54.92	35.51	37.04	41.26
Daugnet	40.94	6.16	0.00	13.20	35.47	0.00	38.69	32.41	21.14	20.38
MCD	14.63	53.40	0.00	9.29	15.67	0.00	45.84	31.46	21.46	32.86
GVBDA	37.53	15.66	**0.06**	7.89	30.64	0.03	32.07	32.70	19.60	23.14
ProDA	39.41	34.20	0.00	16.69	34.49	0.02	49.36	33.78	30.12	33.70
TDARS(ours)	**46.23**	**62.65**	0.00	**21.85**	**36.72**	**0.40**	**64.28**	**38.51**	**47.08**	**50.28**

performance of IoU is less than 10%, the comparison is meaningless.

2) WH2XT: In this case, we use WH as the source city and XT as the target. Compared with CS2XT adaptation shown in Fig. 4, we can see that there exists a larger label distribution divergence on WH2XT, as shown in Fig. 6. The collected intermediate domain holds the same season with the source domain and shares the same geospatial area, sensor, and spatial resolution with the target domain. Hence, the intermediate domain is able to bridge semantic knowledge between source and target domains. Table III depicts the comparison results. We can get an interesting observation from the results that, although DAugNet achieves a better performance than baseline on CS2XT, an undesirable performance can be observed on WH2XT. This mainly results from style transfer that does not consider the semantic relationship between image scenes in different domains, which risks carrying over irrelevant semantic source information to the target domain. In principle, once negative transfer occurs, the more adapted noise is introduced, the worse the model performs because the overall quality of the training data continuously deteriorates. Our TDARS arrives at the best performance. The qualitative comparison results show that traditional UDA methods have a high negative transfer risk in the application of remote sensing images, as shown in Fig. 7. Our method enlarges the relation of the given source and target domains through the selected intermediate domain and performs effective knowledge transfer among all domains. Consequently, the risk of negative transfer has decreased, leading to performance improvements in most categories.

3) WH2YY: We then use WH as the source city and YY as the target. We can see that there exists visual-environment discrepancy and label distribution divergence between the source domain and the target domain, as shown in Fig. 8. The intermediate domain is also selected to hold the same season with the source domain and share the same geospatial area, sensor, and spatial resolution with the target domain. We list the comparison results in Table IV and qualitative results in Fig. 9. We get the same observation holds as on WH2XT adaptation. The proposed TDARS also achieves consistent improvement in most categories.

C. Ablation Studies

1) Ablation Study on the Components in TDARS: We conduct ablation studies on three components of the proposed TDARS: 1) knowledge transfer from positive pseudolabels (given from primary confident class); 2) knowledge transfer from negative labels (given from nonprimary confident class); and 3) the selected intermediate domain.

Adopting the *primary confident class* as *positive pseudolabels* to retrain the network is an intuitive way to reduce the domain gap. We utilize the pretrained source network to generate pseudolabels for target samples. Specifically, for each pixel x_t in a target image, we consider the primary confident class as the pseudolabel on x_t when the highest confidence of SoftMax output is larger than 0.5. Then, we train the target network with pseudolabels on the target domain. However, as shown in Tables V–VII, only utilizing pseudolabels cannot provide a positive adaptation for remote sensing images.

Nonprimary confident classes also contain valuable information. We could give a *negative label* to one class randomly selected from nonprimary classes to prevent the adapted model from overfitting to misclassified data. We utilize the pretrained source network to generate negative labels and positive pseudolabels. After learning through the proposed negative labels, we further train the target network through the filtered target samples with positive pseudolabels. As depicted in Tables V–VII, we observe that utilizing positive pseudolabels together with negative labels increases the possibility of positive adaptation. This also shows the synergy benefit of knowledge transfer via primary and nonprimary confident classes.

The *intermediate domain* keeps a single factor change with the source (target) domain. We perform a transitive adaptation process through this intermediate domain. We first transfer knowledge from the source network to the intermediate domain. Then, we transfer knowledge from the adapted intermediate network to the target domain. Since the intermediate domain has provided more sharing information to bridge the given source and target domains, the domain gap has been reduced by such a transitive adaptation process. As depicted in Tables V–VII, utilizing all three components has an impressive impact on performance improvement.

2) Parameter Sensitivity Analysis of Confidence Threshold: To investigate the effect of the confidence threshold γ in the positive pseudolabel generation, we conduct a series of experiences with γ in the range of $[0.4, 0.8]$. As depicted in Fig. 10(a), the peak performances are obtained when γ is set as 0.5 for all three datasets. The performance is relatively poor when γ is too large (e.g., larger than 0.7). A possible reason is that larger γ may generate fewer pseudolabels that are not enough to retrain the model. We also observe that the proposed method is more sensitive in WH2XT and

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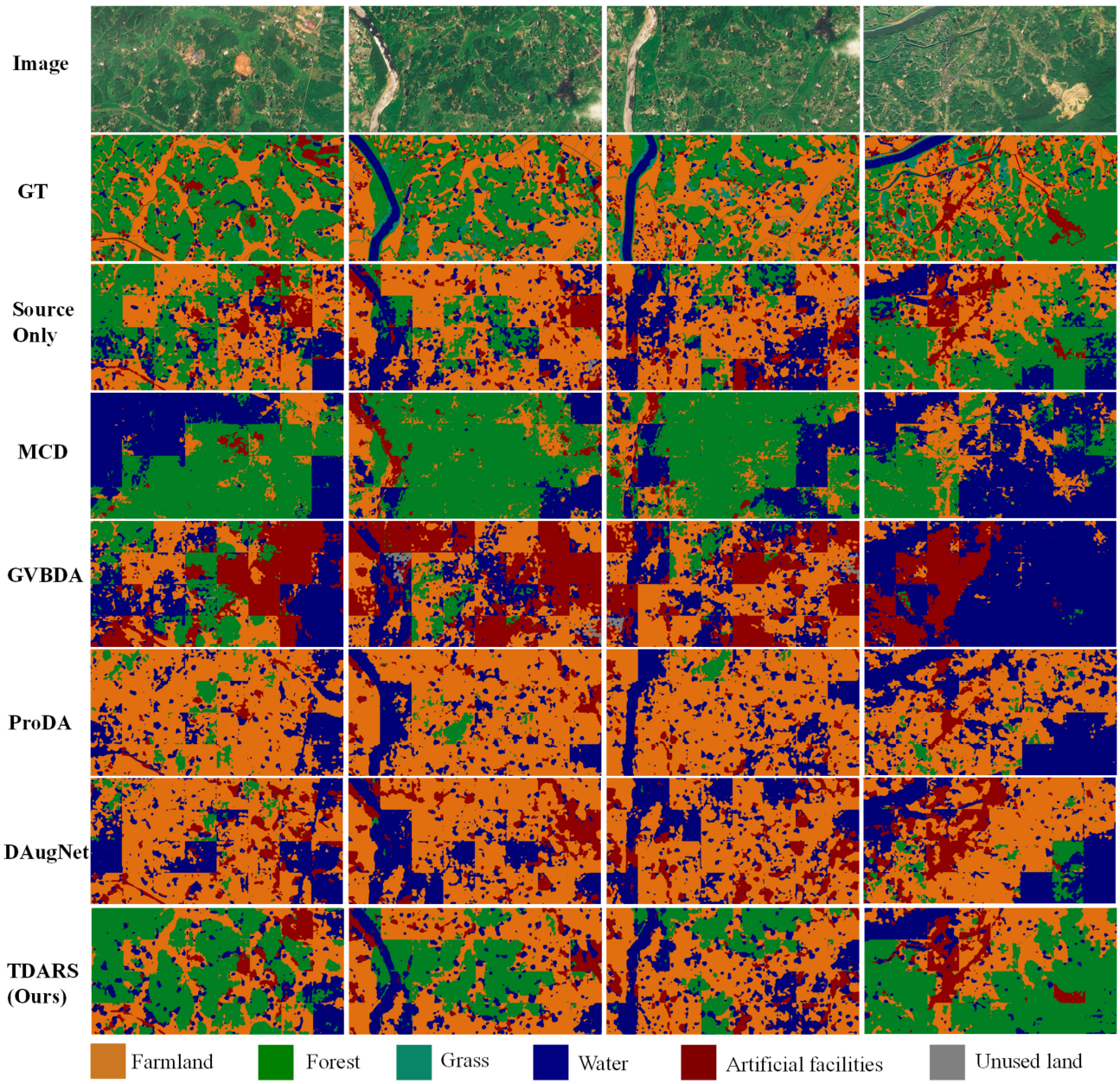


Fig. 7. Qualitative comparisons of adapted segmentation on WH2XT. We splice 32 adapted patches (256 × 256) to a remote sensing image (2048 × 1024).

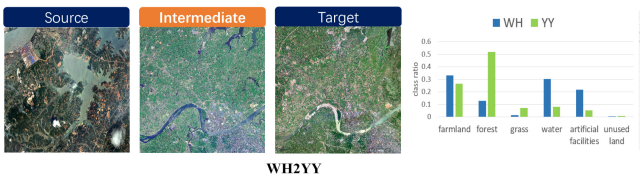


Fig. 8. Remote sensing images (8192 × 8192) sampled from source and target domains of WH2YY and corresponding label distribution of six categories in the source and target domains. The selected image from the intermediate domain holds the same season with the source domain and shares the same geospatial area, sensor, and spatial resolution with the target domain.

WH2YY than that in dataset CS2XT when γ is set larger than 0.5. A conceivable reason is that large γ hardly generates high-confidence pseudolabels during the initial training stages, especially when existing a larger label distribution divergence between the source domain and the target domain, as shown in WH2XT and WH2YY datasets.

3) Parameter Sensitivity Analysis of Balance Weight in Negative Learning: TDARS employs α as a hyperparameter to balance knowledge transfer from less-risk and fuzzy negative labels. Fig. 10(b) exhibits the FWIoU obtained by selecting the hyperparameter α in $\{0, 0.01, 0.02, 0.05, 0.1, 0.2\}$. These results show that the best performance is achieved when α is set as 0.05. When $\alpha = 0$, knowledge transfer only benefits from less-risk negative labels and ignores the information in fuzzy categories. With an increasing α, TDARS gains improvements with joint knowledge transfer from less-risk and fuzzy negative labels. However, the knowledge transfer from less-risk negative labels in (10) does

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TABLE IV

COMPARISON RESULTS (%) OF WH2YY IN TERMS OF IOU FOR EACH CLASS, OA, AA, KAPPA, AND FWIOU. THE BEST IS HIGHLIGHTED IN COLUMN

Method	farmland	forest	grass	water	artificial facilities	unused land	OA	AA	Kappa	FWIoU
src only	46.48	67.17	0.00	24.48	23.64	0.06	62.73	37.91	46.52	50.51
Daugnet	50.66	27.66	0.00	13.11	22.19	0.02	42.87	32.70	28.09	30.06
MCD	31.90	62.79	0.00	15.93	19.42	0.00	54.90	32.13	35.41	43.43
GVBDA	51.08	60.50	0.00	**27.87**	19.55	0.00	61.02	36.44	45.78	48.32
ProDA	**53.59**	64.97	0.00	20.01	20.98	0.00	62.47	36.87	46.05	49.41
TDARS(ours)	49.67	**72.20**	0.00	26.69	**25.44**	**0.07**	**66.16**	**38.52**	**49.85**	**53.59**

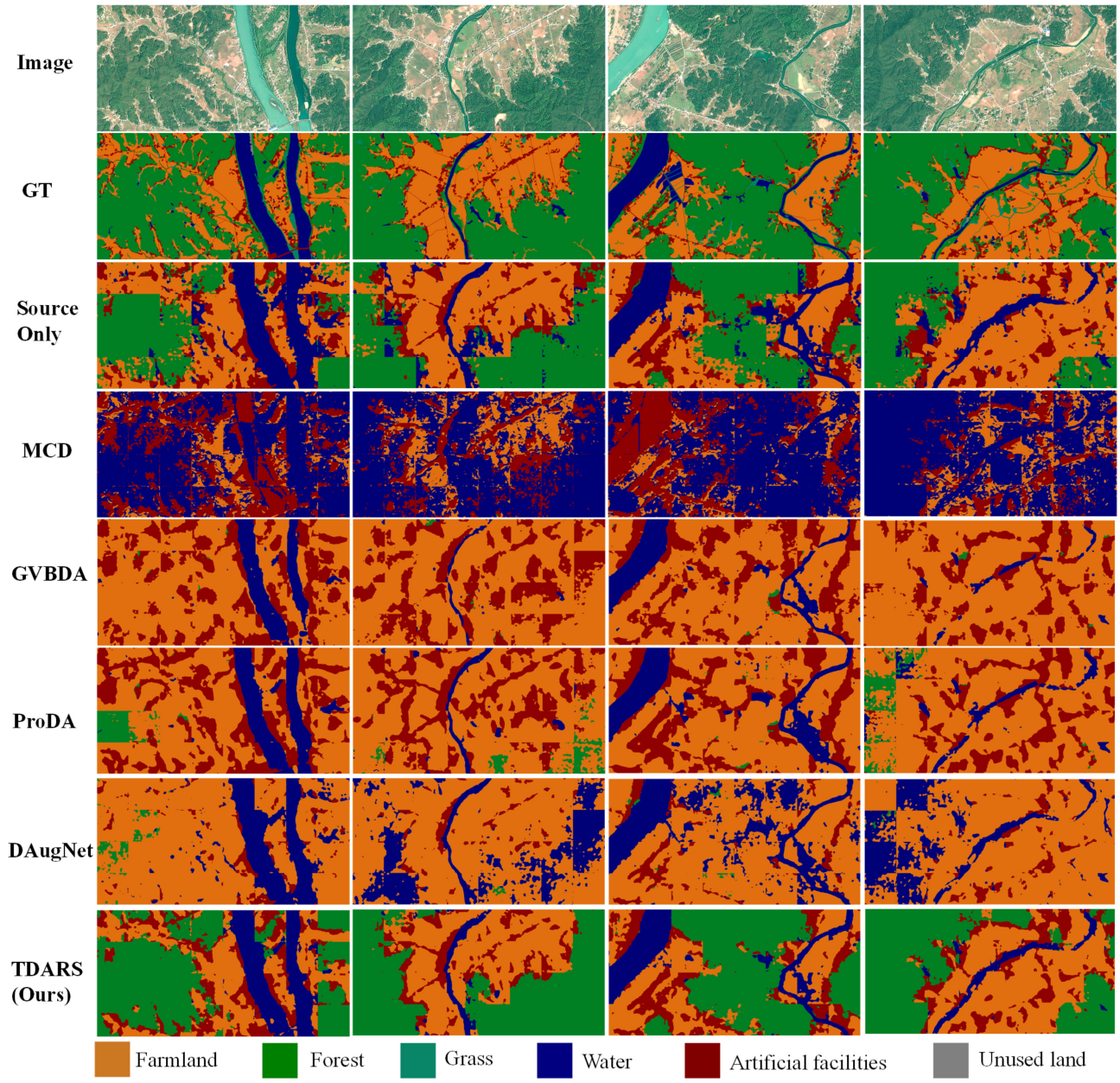


Fig. 9. Qualitative comparisons of adapted segmentation on WH2YY. We splice 32 adapted patches (256 × 256) to a remote sensing image (2048 × 1024).

not dominate weights when α goes too large (e.g., larger than 0.1).

To further understand the impact of less-risk negative labels (rNeg) and fuzzy negative labels (fNeg), we also visualize the confidence percent histogram of target predictions. As shown in Fig. 11, the confidence of correct predictions generated by the nonadapted source model is much lower. By contrast, the confidence of correct predictions is increased after TDARS

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TABLE V
ABLATION STUDY OF EACH COMPONENT IN TDARS ON CS2XT

FWIoU FWIoU-gain	67.32	67.04 0.28-	68.58 1.26+	**70.07** **2.75+**
source	✓	✓	✓	✓
positive pseudo labels		✓	✓	✓
negative labels			✓	✓
Intermediate Domain				✓

TABLE VI
ABLATION STUDY OF EACH COMPONENT IN TDARS ON WH2XT

FWIoU FWIoU-gain	41.26	40.97 0.29-	43.73 2.47+	**50.28** **9.02+**
source	✓	✓	✓	✓
positive pseudo labels		✓	✓	✓
negative labels			✓	✓
Intermediate Domain				✓

TABLE VII
ABLATION STUDY OF EACH COMPONENT IN TDARS ON WH2YY

FWIoU FWIoU-gain	50.51	50.48 0.03-	51.74 1.23+	**53.59** **3.08+**
source	✓	✓	✓	✓
positive pseudo labels		✓	✓	✓
negative labels			✓	✓
Intermediate Domain				✓

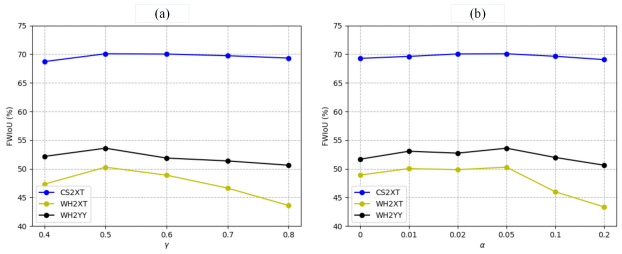


Fig. 10. Sensitivity analysis for parameters used in TDARS. (a) Confidence threshold γ. (b) Balance weight α in negative learning.

(rNeg + Pos). Furthermore, we also observe that the confidence of correct predictions is significantly ahead of error predictions after TDARS (rNeg + fNeg + Pos), which leads to better segmentation performance.

D. Weakness

As demonstrated in previous experimental results, the proposed method performs well in the domain shift from remote sensing images with the aid of the selected intermediate domain. However, the assumption that every single intermediate sample can provide meaningful information as an adaptation bridge may not satisfy. The proposed method fails to assess the contribution of each sample from the intermediate domain. The proposed method also works less well for rare classes, which is also challenging for the other state-of-the-art methods.

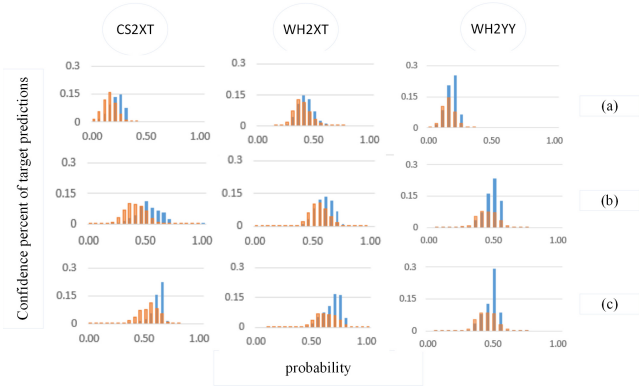


Fig. 11. Confidence percent histogram shows the distribution of target predictions according to probability p_y. Blue indicates correct predictions, whereas orange indicates error predictions. (a) Nonadapted source (CS/WH). (b) TDARS (rNeg + Pos). (c) TDARS (rNeg + fNeg + Pos).

V. CONCLUSION

To overcome the domain shift problem, UDA has received increasing interest in the field of RS image semantic segmentation. An important requirement for successful UDA is that the related information between the source domain and the target domain overrides the irrelevant information, which can be expressed in the similarity of feature or label space. However, the domain shift problem from the RS image is caused by both feature distribution shift and label distribution shift, which are ascribed to VE-changes and SS-changes, respectively. In practical applications, forcibly adapting to a new domain has a high risk of negative transfer. Thus, how to enlarge the relation between the given source and target domains and effectively transfer knowledge among domains attracts much attention. Driven by this urgent demand, this article proposes a transitive domain adaptation method with an introduced intermediate domain for cross-domain RS image semantic segmentation where the intermediate domain is collected to share the visual environment (semantic scene) with the source (target) domain. More specifically, the proposed method learns from primary and nonprimary confident classes to increase the likelihood of transferring valuable information. Extensive experiments under three typical cross-domain settings show that the proposed method outperforms state-of-the-art methods, greatly reducing the risk of negative transfer for cross-domain RS image semantic segmentation. However, the problem of insufficient adaptation learning in extremely rare classes remains unresolved, which is another challenge for cross-domain RS image semantic segmentation. In the future, we plan to further explore the complementary of the proposed method with recent class-balanced techniques to increase the adaptation ability of extremely rare classes. There is room for improvement in the selection of intermediate domains to enlarge the relation between the given source and target domains.

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